

Smart Bus: SMART SAFETY MONITORING SYSTEM FOR LONG DISTANCE BUSES IN SRI LANKA

25-26J-511

Checklist set 1

Git Repository :

<https://github.com/MatheeshaDias0926/Research-Project-25-26J-511-.git>

Wireframe sketches:

SMART BUS PORTAL

Overview

Fleet

Violations

Conductors

Maintenance Log

IOT Simulator

Safety Check

Safety Logic

Scenario Simulator

Profile

Logout

SYSTEM DASHBOARD

47

ACTIVE BUSES

94

TOTAL CONDUCTORS

12

VIOLATIONS

5

MAINTENANCE

SYSTEM STATUS

All Systems Operational

QUICK ACTIONS

MANAGE CONDUCTORS

SAFETY CHECK

VIOLATION ANALYSIS (BAR CHART)

BAR CHART

SAFETY TRENDS (LINE)

LINE CHART

REAL-TIME FLEET CAPACITY (SCATTER)

SCATTER PLOT

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REGISTER NEW CONDUCTOR

USERNAME

PASSWORD

ASSIGN BUS

REGISTER

LIST OF REGISTERED CONDUCTORS

JOHN SMITH

ID: C001

Assigned Bus: ABC-1234

ACTIVE

SARAH JOHNSON

ID: C002

Assigned Bus: DEF-5678

ACTIVE

MIKE DAVIS

ID: C003

Assigned Bus: GHI-9012

OFF-DUTY

EMMA WILSON

ID: C004

Assigned Bus: JKL-3456

ACTIVE

DAVID BROWN

ID: C005

Assigned Bus: MNO-7890

ACTIVE

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FLEET MANAGEMENT

+ ADD

ACTIVE FLEET

NUMBER PLATE	ROUTE	CAPACITY	STATUS	SYSTEM ID
ABC-1234	Route 42	50	Active	SYS-001
DEF-5678	Route 18	45	Active	SYS-002
GHI-9012	Route 7	60	Maintenance	SYS-003
JKL-3456	Route 25	50	Active	SYS-004
MNO-7890	Route 33	55	Active	SYS-005
PQR-2468	Route 12	45	Inactive	SYS-006

SMART BUS PORTAL

IOT SIMULATOR

TELEMETRY CONTROLS

VEHICLE ID

NP-1234

SPEED (km/h)

45

OCCUPANCY

30

VEHICLE STATUS

Clear (Safe)

GPS COORDINATES

LAT

6.9271

LONG

79.8612

SEND DATA PACKET

TRANSMISSION LOG

No packets sent yet.

MAP INTERFACE

SEARCH LOCATION

MAP INTERFACE

(Leaflet/Google Maps)

Lat: 6.9271 | Long: 79.8612

SMART BUS PORTAL

Maintenance Log

MAINTENANCE OVERVIEW

ALL | PENDING | RESOLVED

BUS: NP-1234 | HIGH | REPORTED

BRAKE SYSTEM MALFUNCTION

Front left brake not responding properly

REPORTED BY: CONDUCTOR • 20/12/2025, 20:19:20

BUS: XY-5678 | MEDIUM | IN PROGRESS

ENGINE OVERHEATING

Temperature gauge showing abnormal readings

REPORTED BY: DRIVER • 19/12/2025, 14:30:15

BUS: AB-9012 | LOW | RESOLVED

WINDSHIELD WIPER REPLACEMENT

Rear wiper blade worn out

REPORTED BY: MAINTENANCE STAFF • 18/12/2025, 09:45:00

BUS: CD-3456 | HIGH | PENDING

TRANSMISSION ISSUE

Gear shifting delayed and rough

REPORTED BY: DRIVER • 21/12/2025, 11:20:45

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MY PROFILE

AD

ADMIN USER

SYSTEM ADMINISTRATOR

USER NAME

admin_smartbus_01

ROLE

Portal Administrator

ACCOUNT ID

SB-AUTH-99281

EMAIL ADDRESS

admin@smartbusportal.com

EDIT PROFILE SETTINGS

ACCOUNT INFORMATION

LAST LOGIN:

11 Jan 2026, 14:23

ACCOUNT CREATED:

15 Dec 2025

ACCESS LEVEL:

Full Access

USER

CREATE ACCOUNT

Join as a Passenger

USERNAME

Choose a username

PASSWORD

CONFIRM PASSWORD

REGISTER

Already have an account? [Sign in](#)

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SAFETY & PHYSICS ANALYSIS

RUN A REAL-WORLD SIMULATION USING THE LIVE PHYSICS ENGINE

1. SIMULATION PARAMETERS

SEATED PASSENGERS

STANDING PASSENGERS

SPEED (km/h)

GPS COORDINATES

RUN SAFETY ANALYSIS

MAP INTERFACE

Select Location

Lat: 6.8771 (Jayanagar)

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LIVE SCENARIO SIMULATOR

DIGITAL TWIN SIMULATION POWERED BY ML & REAL-TIME TELEMETRY

1. BUS & ENVIRONMENT

SPEED ZONE

1 50

SPEED

1 40

STOPPING

1 20

BUS SELECTION

1 NP-1234 (R. ROUTE-158)

2. ROUTE PLANNER

START (0800) 1 1 (0900)

1 1 (0900)

CALCULATE PATH

START LIVE SIMULATION

SEARCH

[INTERACTIVE MAP PLACEHOLDER]

Bus: NP-1234 | Speed: 55 km/h
Passengers: 50 (40 seated, 10 standing)

LOGO

WELCOME BACK

Sign into your account

Username

Password

SIGN IN

Don't have an account? [Register](#)

SMART BUS PORTAL

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VIOLATION LOG

SELECT BUS: ALL BUSES

NUMBER PLATE	VIOLATION TYPE	LOCATION	TIME
XYZ-1234	Speeding (85km/h in 60km/h zone)	Main Street, Sector 4	2026-01-10 14:35
ABC-5678	Unauthorized Route Detour	Park Avenue Junction	2026-01-10 09:12
DEF-9012	Overcapacity (58/50 passengers)	Central Station	2026-01-09 17:45
GHI-3456	Missed Scheduled Stop	North Terminal	2026-01-09 12:20
ADDITIONAL LOG ENTRIES WILL APPEAR HERE...			

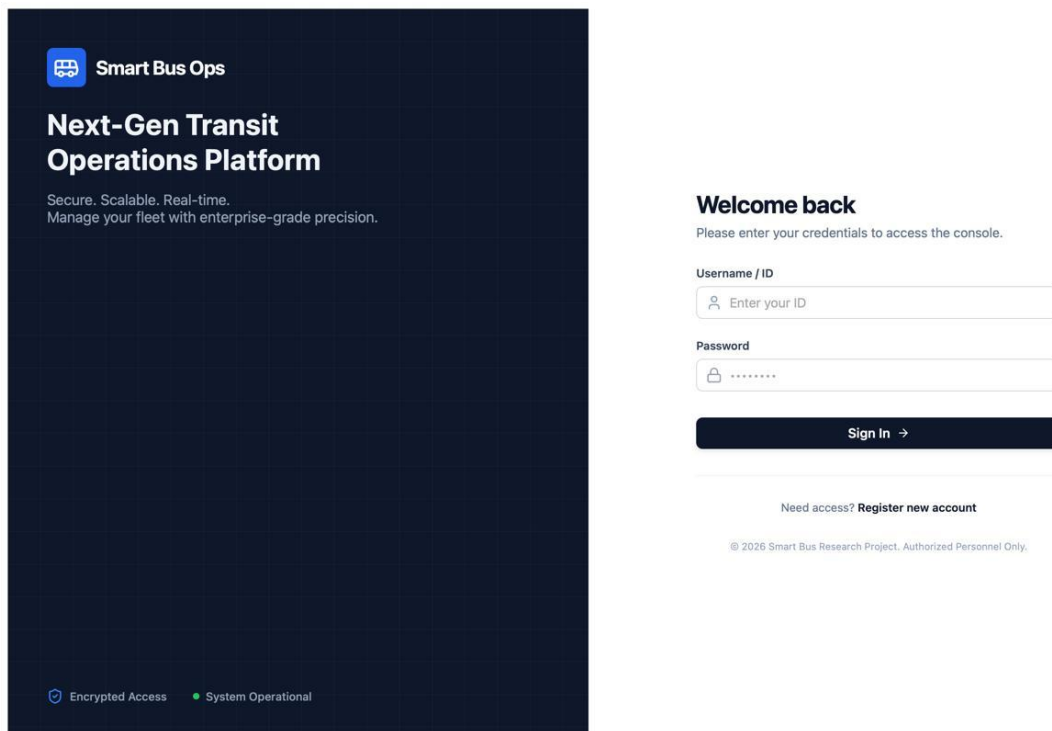
User Feedback :

- The login page needs a more official and authoritative appearance.
- On the dashboard, it is difficult to clearly distinguish between urgent and non-urgent information.
- Tables should provide options to modify or update the displayed data.
- The “Active Crash Notification” does not stand out enough and should be more prominent.
- Attach the captured image of the violation to the authority violation log for visual verification.
- Clearly display the violation type in the violation log for easy identification.

Improvements of UI:

- Enhanced the login page with a professional gradient design and clear authority-oriented subtitles to improve credibility and trust.
- Improved dashboard clarity by introducing color-coded cards (Red, Orange, Green) for quick identification of urgency levels.
- Updated tables to include status badges along with clearly differentiated Edit, Delete, and Change Status action buttons.
- Increased the visibility of the active crash notification by applying a red background to draw immediate attention.

UI mockups :



SmartBus

Passenger Portal

Dashboard

Prediction

Profile

Logout

Live Bus Tracker

Real-time updates on occupancy, speed, and safety.

3 Active Buses

NP-1234

Live

Route ROUTE-138

Crowding

High (60/55)

Speed

60 km/h

Safety

Normal

Tracking Active

View full details →

WP-5678

Live

Route ROUTE-138

Crowding

High (75/50)

Speed

55 km/h

Safety

Normal

Tracking Active

View full details →

NP-5644

Live

Route ROUTE-140

Crowding

High (88/45)

Speed

60 km/h

Safety

Normal

Tracking Active

View full details →

Open "localhost:5173/passenger/prediction" in a new tab

SmartBus

Conductor Portal

My Bus

Maintenance

Profile

Logout

My Bus Dashboard

Real-time overview of your vehicle status.

Refresh Report Issue

NP-1234

Panangala Dhahe kade

Route ROUTE-138

Speed

60 km/h

Passengers

60 / 55

Status

Active

Recent Alerts

overcrowding

11:20:22 AM

Location: 6.2553, 80.3267. Speed: 60 km/h

overcrowding

11:20:22 AM

Location: 6.2553, 80.3267. Speed: 60 km/h

overcrowding

11:20:22 AM

Location: 6.2553, 80.3267. Speed: 60 km/h

overcrowding

11:20:22 AM

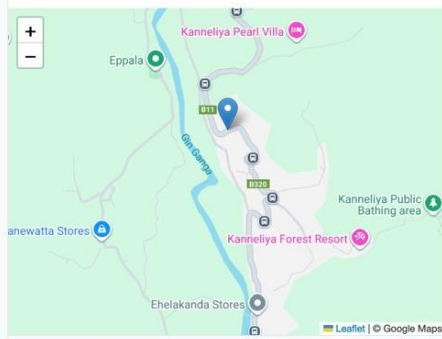
Location: 6.2553, 80.3267. Speed: 60 km/h

overcrowding

11:20:21 AM

Location: 6.2554, 80.3268. Speed: 60 km/h

Live Location



Updated: 7:33:51 PM

Signal: Strong

SmartBus

Conductor Portal

My Bus

Maintenance

Profile

Logout

Report Issue

Submit a maintenance request for your assigned bus.

Maintenance Ticket

REPORTING FOR BUS

NP-1234

Route ROUTE-138 • System ID: 691978294f5541d466caa7e0

Issue Title

e.g. Engine Overheating, Brake Noise

Detailed Description

Describe the problem in detail so the maintenance team can understand...

Priority Level

Low - Minor cosmetic issues

Submit Maintenance Request

SmartBus

Authority Portal

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System Overview

Active Buses

3

Total Conductors

2

Violations (24h)

0

Maintenance

2

System Status

Healthy

Quick Actions

Manage Conductors

Register & Assign Buses

Safety Check

Run Physics Simulation

Violation Analytics (Top Offenders)

SmartBus

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Violation Log

Select Bus: NP-1234 (ROUTE-138)

NP-1234

Overcrowding

Violation type: overcrowding

Speed: 60 km/h

Udagama-Hiniduma Road, Panangala Dhahe kade, Galle District, Southern Province, Sri Lanka

11:20:22 AM

View on Google Maps

1/8/2026

NP-1234

Overcrowding

Violation type: overcrowding

Speed: 60 km/h

Udagama-Hiniduma Road, Panangala Dhahe kade, Galle District, Southern Province, Sri Lanka

11:20:22 AM

View on Google Maps

1/8/2026

NP-1234

Overcrowding

Violation type: overcrowding

Speed: 60 km/h

Loading location...

11:20:22 AM

View on Google Maps

1/8/2026

NP-1234

Overcrowding

Violation type: overcrowding

Speed: 60 km/h

Loading location...

11:20:22 AM

View on Google Maps

1/8/2026

SmartBus

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Fleet Management

+ Add New Bus

Active Fleet

LICENSE PLATE	ROUTE	CAPACITY	STATUS	SYSTEM ID
NP-1234	ROUTE-138	55	Active	691978294f5541d466eaa7e0
WP-5678	ROUTE-138	50	Active	691978294f5541d466eaa7e1
NP-5644	ROUTE-140	45	Active	6946b2fc6da8cdb6b5cc3545

SmartBus

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Manage Conductors

Register new conductors and manage existing fleet assignments.

Register New Conductor

Username

e.g. johndoe

Password

Min. 6 characters

Assign Bus 1 buses available

Select a Bus to Assign

Register Conductor

Registered Conductors 2

Search by name or bus number...

C

Conductor

NP-1234 • Route ROUTE-138

Active

C

Conductor1

WP-5678 • Route ROUTE-138

Active

SmartBus

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Maintenance Overview

All Pending Resolved

Bus: NP-1234 MEDIUM Reported

test

test

Reported by: Conductor • 12/20/2025, 9:24:31 PM

Bus: NP-1234 HIGH Reported

Brake system malfunction

Front left brake not responding properly

Reported by: Conductor • 12/20/2025, 8:19:20 PM


SmartBus

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 **IoT Simulator**

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IoT Simulator

Telemetry Controls

License Plate

Speed (km/h)

Occupancy

Footboard Status

GPS Coordinates

Lat

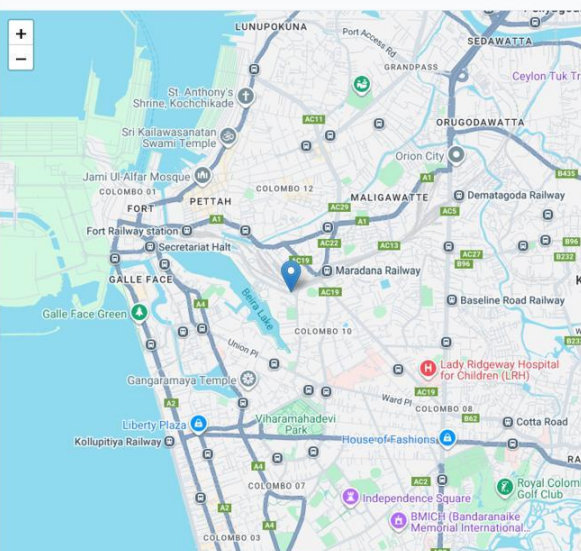
Lon

Send Data Packet

Transmission Log

No packets sent yet.

Select Location




SmartBus

Authority Portal

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Safety & Physics Analysis

Run a real-world simulation using the live physics engine. Enter coordinates manually or select a location on the map.


1. Simulation Parameters

Seated Passengers

Standing Passengers

Speed (km/h)

Location Coordinates

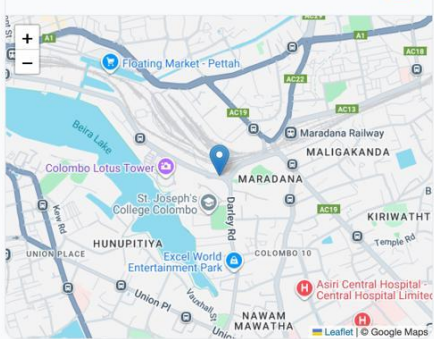
Latitude

Longitude

* Search location or click on the map to auto-fill.

Run Safety Analysis

Select Location



 **SmartBus**
Authority Portal

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Safety Logic & Theories

Comprehensive documentation of the physics models and algorithms driving the Smart Bus Safety System.

- Rollover Prediction
- Stopping Distance
- Road Geometry
- Constants & Thresholds
- Real-time AI Integration
- Simulation Architecture
- Model Verification
- Model Training Details

Rollover Prediction Model

The core of our safety system is the **Digital Twin** model for the Ashok Leyland Viking bus. It predicts rollover risk in real-time by comparing the vehicle's **Lateral Acceleration** against its **Static Stability Factor (SSF)**.

1. Static Stability Factor (SSF)

Resistance to rollover based on geometry. Lower is worse.

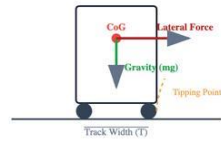
$$SSF = T / (2 \times h_{CoG})$$

2. Lateral Acceleration

The "tipping force" felt while cornering.

$$a_{lat} = v^2 / (r \times g)$$

Force Vector Diagram



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Live Scenario Simulator

Digital Twin Simulation powered by ML & Real-time Telemetry.

Model: XGBoost-Hybrid Latency: ~20ms

1. Bus & Environment

Speed (km/h)
60

Seated: 40 Standing: 20

Road Condition
Dry Asphalt (Safe)

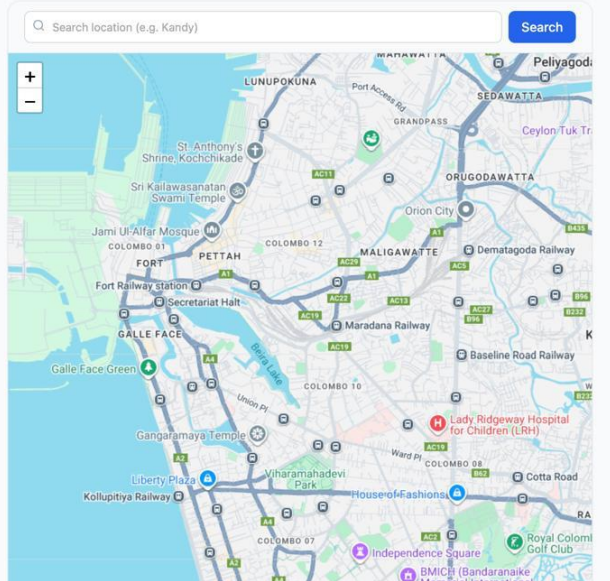
Bus Selection

Select Real Bus
NP-1234 (R: ROUTE-138)

Safety data will be sent to this bus ID in the backend.

2. Route Planner

Start (Green) End (Red)



SmartBus

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Road Condition

Wet / Raining (Risk)

Bus Selection

Select Real Bus

NP-1234 (R: ROUTE-138)

Safety data will be sent to this bus ID in the backend.

2. Route Planner

Start (Green) End (Red)

Calculate Path

Stop

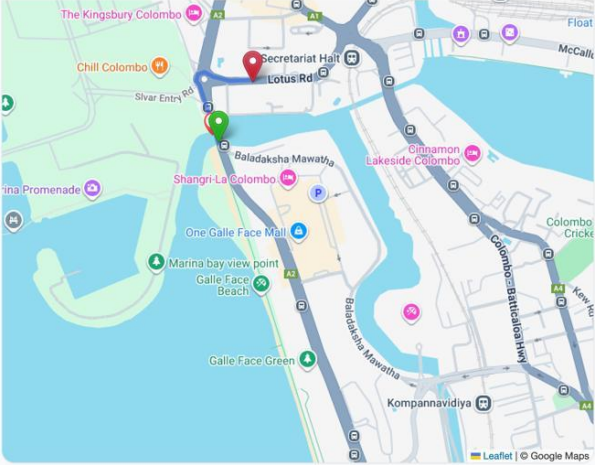
PREDICTED DRIVER ALERT

CRITICAL (1.75)

Warning sent ahead of curve

Curve Ahead: 127 meters

Stopping Dist: 70.5 m



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2. Route Planner

Start (Green) End (Red)

Calculate Path

Stop

DRIVER ALERT LEVEL

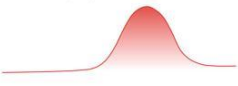
SAFE (0.19)

Stopping Dist: 70.3 m

5-SEC FORECAST

Upcoming Risk: SAFE

RISK TREND (Live)



INCIDENT LOG

7:36:10 PM - Critical Risk Detected (Score: 1.29)

Speed: 60 km/h

7:36:09 PM - Critical Risk Detected (Score: 1.69)

Speed: 60 km/h

7:36:08 PM - Critical Risk Detected (Score: 1.76)

Speed: 60 km/h (LOOKAHEAD)



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2. Route Planner

Start (Green)

End (Red)

Calculate Path

Stop

DRIVER ALERT LEVEL

SAFE (0.19)

Stopping Dist:

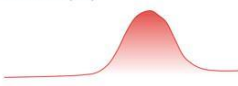
70.3 m

5-SEC FORECAST

Upcoming Risk:

SAFE

RISK TREND (Live)



INCIDENT LOG

7:36:10 PM - Critical Risk Detected (Score: 1.29)

Speed: 60 km/h

7:36:09 PM - Critical Risk Detected (Score: 1.69)

Speed: 60 km/h

7:36:08 PM - Critical Risk Detected (Score: 1.76)

Speed: 60 km/h (LOOKAHEAD)



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Bus Selection

Select Real Bus

NP-1234 (R: ROUTE-138)

Safety data will be sent to this bus ID in the bus

2. Route Planner

Start (Green)

End (Red)

Calculate Path

Stop

PREDICTED DRIVER ALERT

CRITICAL (1.76)

Dashboard

My Bus Dashboard

Real-time overview

CRITICAL RISK

Stability: 0%

Curve Ahead: 126 m

SLOW DOWN IMMEDIATELY

NP-1234

Colombo

Route ROUTE-138

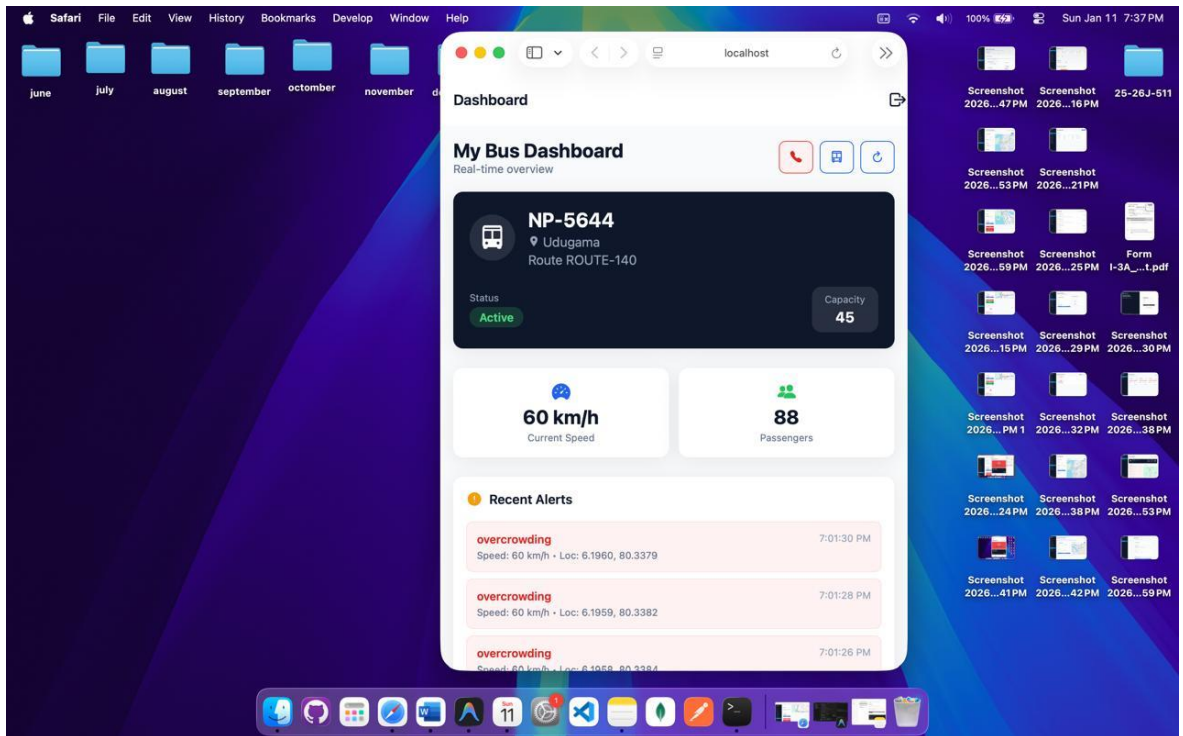
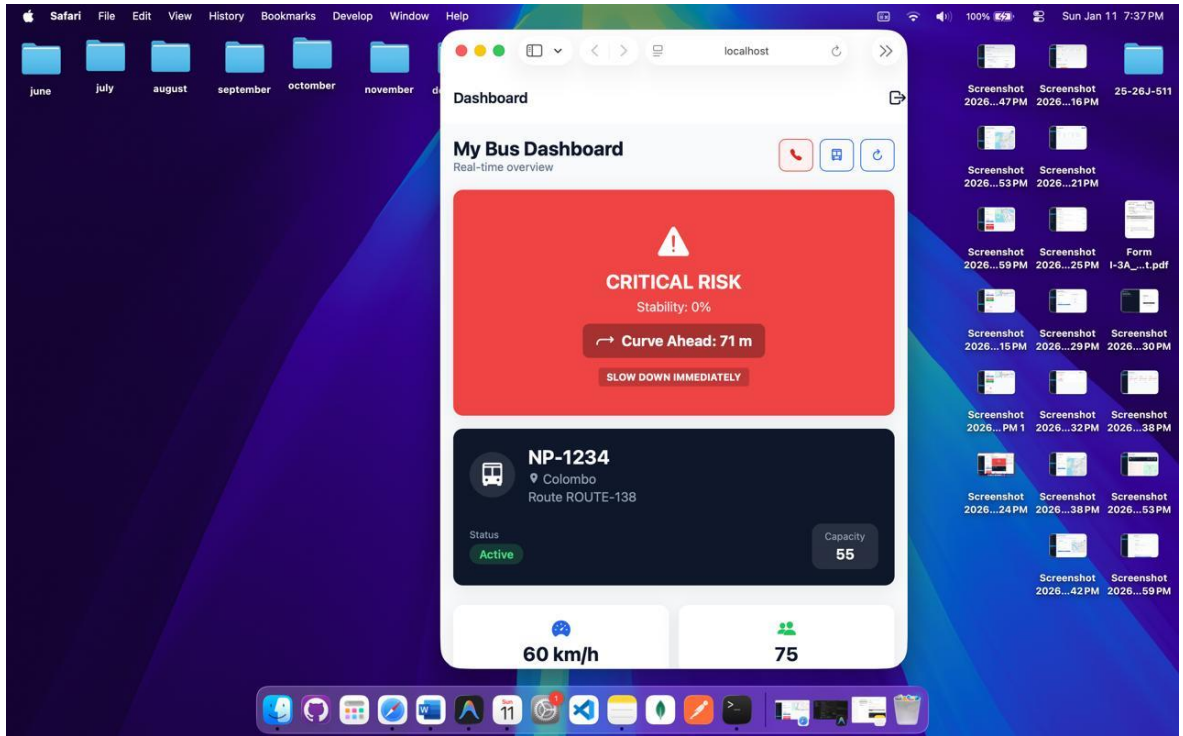
Status: Active

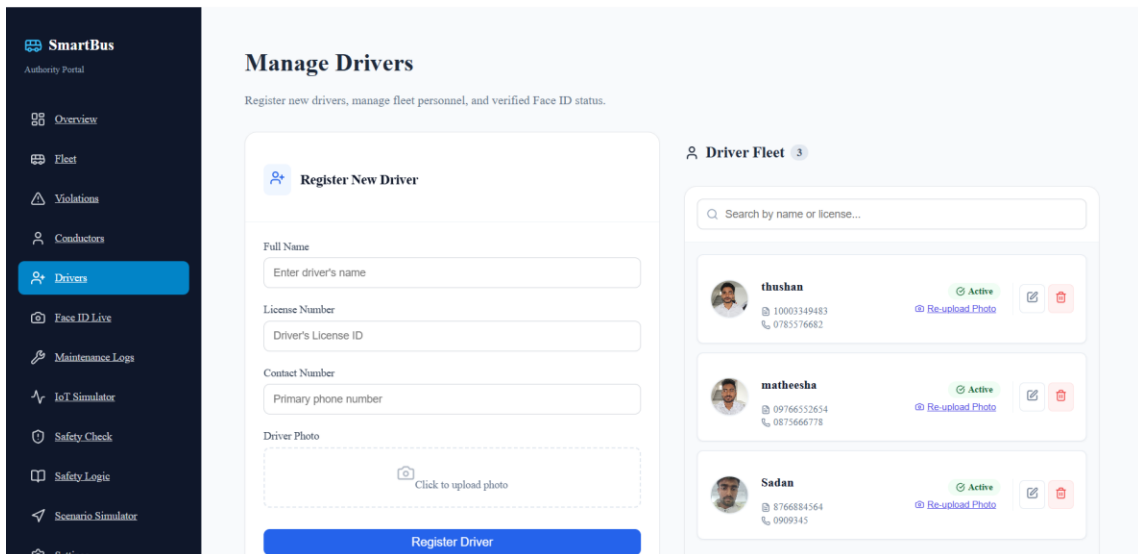
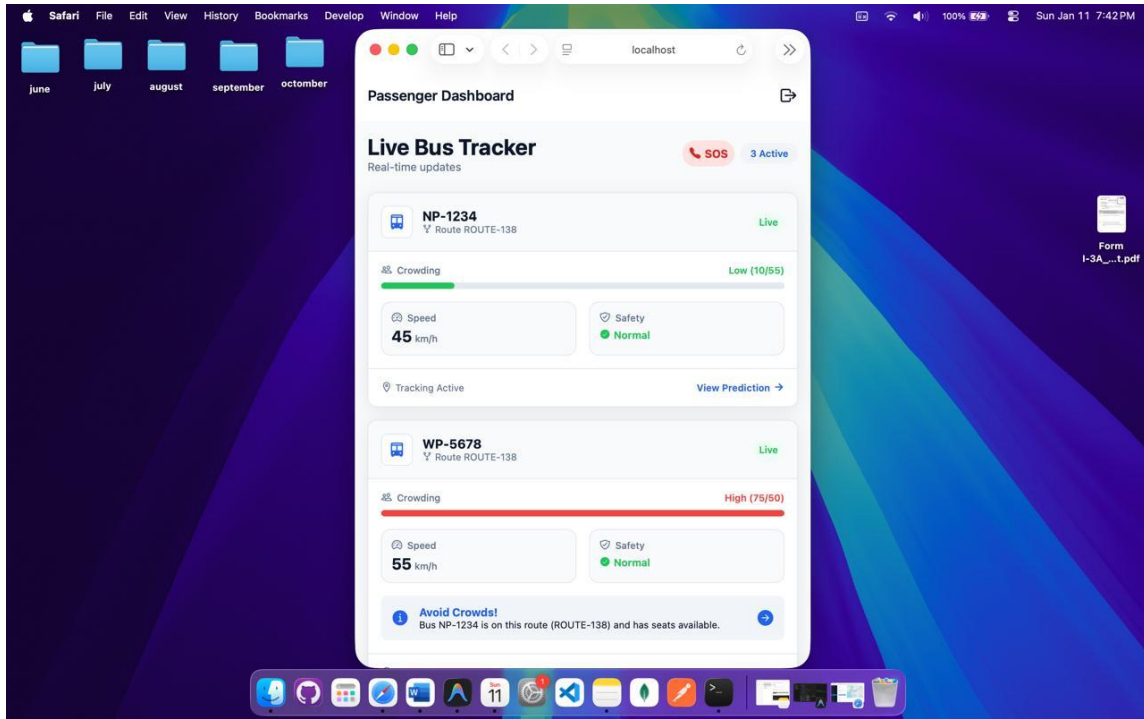
Capacity: 55

60 km/h

75









00 Overview

 [Fleet](#)

 Violations

 Conductors

 Drivers

 Face ID Live Maintenance Logs

IoT Simulator

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Scenario Simulator

20 Settings

Smart Bus Biometric Control Center

AUTO **MANUAL**

Unknown Driver
NOT VERIFIED

ONLINE 0

```

TERMINAL
> System initialized

```

Crash Management System

ADMIN Admin User Logout

Dashboard

Crashes

Buses

Bus Owners

Drivers

Conductors

Police Stations

Hospitals

Settings

Crash Management

All Crashes ▾

BUS ID	TIMESTAMP	LOCATION	SEVERITY	STATUS	MAX ACCELERATION	DETAILS
No crashes found						